

CS253 - Data Structures and Algorithms for DS/AI

Spring 2024

PSO 3, Week 4

Question 1

(Linked List) Consider a sorted circular doubly linked list of N numbers where the head element points to the smallest element in the list. Provide the asymptotic complexity in big- Θ with a brief explanation (including assumptions and analysis for each case, if there is more than one) for the following operations.

1. Inserting an element in its sorted position.
2. Finding the smallest element in the list.
3. Finding the 3^{rd} - largest element in the list.
4. Finding the median in the list.

Question 2

Design a simple browser history feature that allows a user to visit websites, go back in history a certain number of steps, and go forward in history a certain number of steps. The browser history should be implemented using a doubly linked list to efficiently manage the history of visited websites.

Features to Implement:

- Visit: When a user visits a new website, add the website to the browser history. If the user goes back and visits a new website, clear the forward history.
- Back: Given a number steps, enable the user to go back in history by steps number of websites. If steps exceeds the number of websites available in the history, go back to the first visited website.
- Forward: Given a number steps, enable the user to go forward in history by steps number of websites, assuming they have previously gone back. If steps exceeds the available forward history, go to the most recent website.

Don't forget to do the respective analysis.

Question 3**(Binary Tree)**

- (1) A full binary tree cannot have which of the following number of nodes?
- A. 3
 - B. 7
 - C. 11
 - D. 12
 - E. 15
- (2) Given the number of nodes $n = 7$, how many distinct shapes can a full binary tree have?
- A. 3
 - B. 4
 - C. 5
 - D. 6
 - E. 7
- (3) The number of leaf nodes is always greater than the number of internal nodes in a full binary tree.
- A. True
 - B. False
- (4) The number of leaf nodes is always greater than the number of internal nodes in a complete binary tree.
- A. True
 - B. False
- (5) Given the number of nodes in a full binary tree, the number of its leaf nodes is determined.
- A. True
 - B. False